

AI-driven Data Analytics for Cyber Threat Intelligence and Anomaly Detection

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Abstract: Cyber threats are evolving rapidly, making traditional malware detection approaches insufficient in handling sophisticated attacks such as zero-day exploits, polymorphic malware, and advanced persistent threats. As a solution to these problems, this study presents a novel hybrid AI-based data analysis framework for cyber threat intelligence and anomaly detection. We specifically proposed a model that combines autoencoders, random forests, and CNN-LSTM architectures to improve anomaly detection and classification. The dataset used for training and evaluation is the Microsoft Malware Dataset, a publicly available dataset. The AE-RF module learns the normal behavior of the system to detect abnormal behavior, and the CNN-LSTM model identifies spatial and temporal properties of malware for execution patterns. This article ensures that the dataset is well-preprocessed, including feature extraction, PCA, and SMOTE for class balancing to provide optimal model performance. The experimental results show that AE-RF has an area under curve (AUC) score of 0.91 and CNN-LSTM has 0.94, which demonstrates that unsupervised and supervised techniques together have high classification accuracy. This hybrid approach not only improves malware detection accuracy but also enhances cyber threat intelligence by providing interpretable insights into attack patterns. The study concludes that integrating multiple AI techniques can significantly improve cybersecurity defenses against evolving threats, paving the way for real-time, AI-powered malware detection solutions.

Keywords: *Cyber Threat Intelligence, Anomaly Detection, Autoencoder-Random Forest (AE-RF), CNN-LSTM Model, Malware Detection*

I. INTRODUCTION

Cyber threats have gotten more sophisticated with criminals using advanced techniques such as polymorphic malware, ransomware, and zero-day exploits [1]. Traditional security controls that use mostly signature-based and heuristic-based detection methods cannot keep pace with the sheer speed of new malware variants. Classical methods work based on heuristics and known attack signatures [2], making them susceptible to new or encrypted attacks. Due to this fact, AI-powered data analytics is now an esteemed approach that may assist in enhancing cyber threat intelligence by utilizing the implementation of machine learning and deep learning models

for anomaly detection as well as malware classification. Machine learning algorithms, especially unsupervised learning algorithms like autoencoders, are suitable for the detection of anomalies [3]. Autoencoders learn the typical behavior of a system and are able to detect deviations, thus being good at detecting hitherto unknown malware variants. However, an autoencoder will not provide as precise a classification. To overcome this limitation, this framework employs random forest classifiers to further analyze the autoencoder's extracted anomalies to identify whether the detected anomalies are benign or malicious. Additionally, the sequential behavior of malware can be modeled as the optimal solution to be run through deep learning models. CNN- LSTM models [4] perform excellently for that. Convolutional Neural Network (CNN) learns spatial patterns from unprocessed byte streams and opcode commands, whereas Long Short-Term Memory (LSTM) network learns temporal relationships, enabling us to identify variants of malware that mutate [5].

This study suggests a hybrid anomaly detection and classification system combining autoencoders, random forests, and CNN-LSTM networks. A Microsoft Malware Dataset, a publicly released malware classification dataset, is employed to train and test the method described in the current study. Thus, the primary objective is to enhance the precision of malware detection and give an understandable interpretation of cyber threats to facilitate security analysts to react more effectively towards the more sophisticated nature of cyber threats.

II. BACKGROUND STUDY

Over the years, numerous studies have explored the application of machine learning and deep learning techniques in cybersecurity. Static and dynamic file analysis methods have traditionally been used in rule-based approaches to malware detection. Static analysis looks at file structures, opcode sequences, API calls, etc, meanwhile, dynamic analysis views the actual execution behavior in a sandbox. Nevertheless, the techniques described in these approaches are computationally expensive and are mined against evasion strategies. Autoencoders have been widely studied for unsupervised anomaly detection. They work by encoding input data into a

lower-dimensional representation and reconstructing it. The reconstruction error acts as a measure of anomaly, as malware samples tend to exhibit higher reconstruction errors compared to normal files. Deep autoencoders with multiple layers have demonstrated superior performance in learning complex data distributions, making them useful in identifying malware that deviates from normal system behavior. L. Zygal *et al* [6] discussed an autoencoder-based anomaly detection system for the CMS electromagnetic calorimeter, utilizing a Residual Neural Network architecture. It efficiently detects anomalies by measuring reconstruction loss, achieving low false discovery rates while identifying detector issues in real time. 100,000 good images processed offline for training. The validation set includes good images with 'fake' anomalies. Autoencoder detects anomalies missed by existing systems. A low false discovery rate is achieved in anomaly detection. Random forest is an ensemble learning method used for malware classification and feature importance analysis. The random forest is an ensemble learning method that constructs a multitude of called decision trees at training time and outputs the class that is the mode of the classes output by individual trees. Random forests have shown great performance in malware detection tasks due to their ability to manage high-dimensional data, resilience to noise, and explainability, outperforming many traditional classifiers. The Random Forest (RF) Ensemble method [7] excels in malware classification by combining multiple decision trees, enhancing robustness and accuracy. The network was designed to be superior to the data classifiers commonly used to detect obfuscated malware because of its ability to handle high-dimensional data and reduce overfitting. Malicious Memory Dumps Obfuscated-MalMem2022 dataset. Sources for various Ransomware, Spyware, and Trojan Horses. Model detection accuracy of over 99%. The Gradient Boosting ensemble works excellently on malware classification. The study [8] found that the Random Forest classifier achieved 100% accuracy in malware detection, demonstrating its effectiveness in handling high-dimensional data and providing robust predictions. This aligns with findings that Random Forests outperform traditional classifiers in cybersecurity applications. Dataset from Kaggle library with 373 samples, 301 malicious. 531 characteristics extracted from Windows programs. RF, SGD, extra trees, and Gaussian NB classifiers achieved 100% accuracy, precision, recall, and f1-score in the test. Autonomous behavior-based malware analysis and machine learning can effectively identify malware. A hybrid CNN-LSTM approach allows for a deeper understanding of how malware evolves, improving detection rates for polymorphic and metamorphic malware variants. Combining multiple AI techniques has been shown to improve the robustness and accuracy of malware detection systems. Prior studies have integrated autoencoders with supervised classifiers to enhance anomaly detection, while CNN-LSTM architectures have been used for analyzing malware behavior in execution logs. The paper [9] focuses on enhancing XSS attack detection by combining Universal Sentence Encoder and Word2Vec embeddings with machine learning and deep learning techniques rather than integrating autoencoders or CNN-LSTM architectures for malware detection, as described in your question. XSS dataset from Kaggle by Syed Saqlain Hussain Shah. 13,685 entries from PortSwigger and OWASP Cheat Sheets. The combined embeddings approach outperforms traditional methods in detection. Superior accuracy, precision, recall, ROC, and F1-score achieved.

III. PROPOSED METHODOLOGY

A. Dataset

The Microsoft Malware Dataset [10] is a widely recognized benchmark dataset that is extensively used in malware detection research. It contains a wide range of malicious software samples and allows you to analyze and classify the malware based on various ML approaches. However, it is highly useful because it contains static analysis features from various malware families, which can help identify the pattern and behavior of a certain malware. Unlike dynamic analysis, which requires executing the malware in a controlled environment, static analysis leverages pre-extracted features from executable files, making it efficient and scalable for large-scale detection systems. The dataset includes several key feature types that contribute to robust malware classification and anomaly detection. Byte-level representation offers raw hex sequences directly obtained from malware binaries, granting an insight into the low-level representation of file structure and content. Feels natural to utilize these translated into a numerical feature vector representation and have models learn some pattern that signifies bad behavior. Another significant feature type is opcode sequences, which are the instruction sets malware runs. These sequences are encoded features with TF-IDF (Term Frequency-Inverse Document Frequency), which emphasizes the most important opcodes in distinguishing benign from malicious software. In addition, the dataset also contains metadata features such as file size, entropy, and timestamps that offer information regarding the nature of malware. Entropy, for instance, is a measure of randomness in a file, which can be helpful in detecting packed or obfuscated malware. Such as file timestamps and size, which can simplify classification by showing trends in the evolution and distribution of malware.

The goal of using this dataset is to come up with meaningful attributes to improve malware classifying accuracy. Taking advantage of these three Pillars, a Complete Hybrid Model Our suggested hybrid model is based on three essential principles: byte-level analysis, opcode frequency encoding, and metadata features, combined in harmony to develop an effective solution for identifying and classifying potential malware.

B. Data Preprocessing

Data preprocessing is a fundamental step in preparing the Microsoft Malware Dataset for machine learning-based malware detection. Due to the high dimensionality of the dataset and its uneven class distributions, it undergoes several transformation processes to ensure that the model is trained optimally. Preprocessing sinks into feature extraction and selection, which in the group used to that the unstructured mal was taken samples and represent them in a struct that uses numerical way. These are tokenized and vectorized using TF-IDF (Term Frequency-Inverse Document Frequency) encoding to process opcode sequences that can expose the actual instruction sets being executed by the malware. In doing so, this method quantifies the importance of each opcode in a malware sample, and as a result, the model concentrates on its distinguishing facets. Byte sequences, derived from malware executables, are transformed into fixed-length feature vectors, providing a standardized input format for all samples. Furthermore, file metadata features (e.g., file entropy, size, timestamps) are normalized to provide consistent scaling and

avoid having certain attributes overshadow other learning processes.

For dimensionality reduction, Principal Component Analysis (PCA) is used considering the high dimensionality of the extracted features. This method discards redundant or less relevant features, and keeps only the most relevant components from the dataset, resulting in increased computational performance without loss of classification accuracy. Since malware datasets often exhibit severe class imbalances where some malware families are overrepresented while others have very few samples Synthetic Minority Over-sampling Technique (SMOTE) is applied to generate synthetic samples for

underrepresented classes. By doing this resampling, we avoid the model overfitting the majority class, also it allows the model to generalize for different malware families.

These preprocessing steps are applied to the dataset as shown in Figure 1 by feature extraction, dimensionality reduction, normalization, and balancing the class of images in the dataset which shapes the dataset into a structured and optimized form. This guarantees that machine learning models later in the pipeline will learn from the data properly which makes the malware detection performance better with significantly reduced overhead.

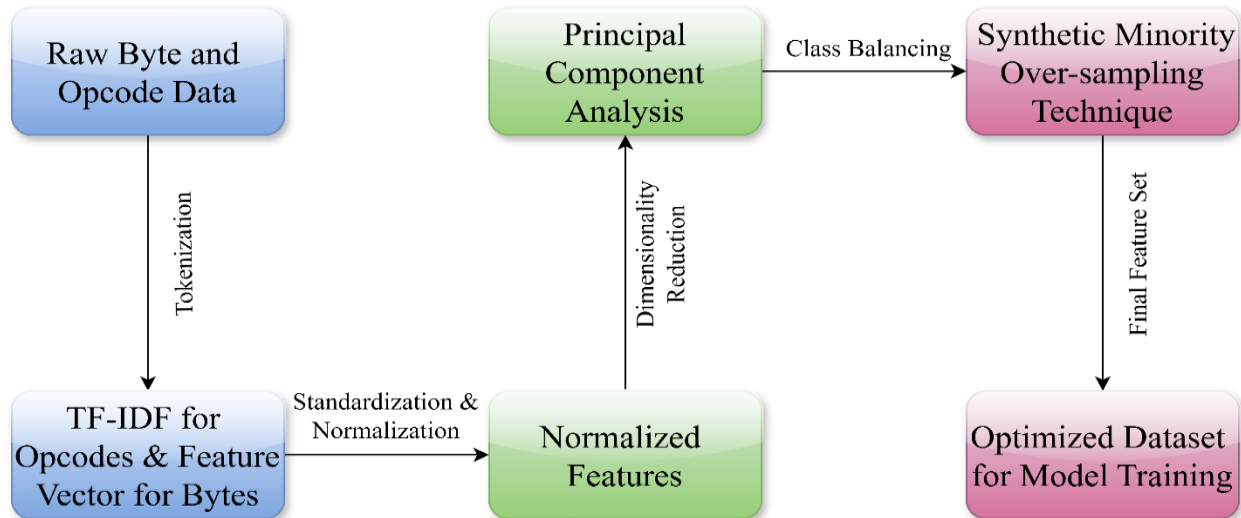


Fig. 1. Data Preprocessing Workflow

C. Architecture of the Model

The proposed hybrid model integrates Autoencoder-Random Forest (AE-RF) and CNN-LSTM architectures to leverage the strengths of both anomaly detection and supervised classification for malware detection. The blending of these two strategies with feature reconstruction and sequential pattern recognition improves the model's potential to detect known and unknown malware.

It implements its anomaly detection module through the Autoencoder-Random Forest (AE-RF), which learns the benign activity of the files. Autoencoders are a class of unsupervised neural networks that encode the patterns of benign samples to minimize reconstruction error. When a new sample is processed, the autoencoder attempts to reconstruct it. If the reconstruction error is significantly high, it indicates that the sample deviates from normal behavior and is likely malicious. To further refine this anomaly detection process, the reconstruction errors are passed to a Random Forest classifier, which determines whether the anomaly corresponds to a known malware family or a novel attack. Because the random forest model is composed of many decision trees, the output of the model is determined based on the anomaly scores of benign and malware samples that were used to train the model, making it a strong candidate to detect an outlier as malicious behavior. Likewise, we propose a CNN-

LSTM module for analyzing sequential tensors taken from malware programs to detect advanced classes of malware that evolve. CNN (Convolutional Neural Network) works as a feature extractor from byte sequences of malware samples to detect spatial patterns. Such spatial representations are also effective at identifying localized patterns (e.g., opcode frames or encoded payloads) present in malware binaries. Then, the extracted features are fed into an LSTM (Long Short-Term Memory) network to capture temporal dependencies in the malware's execution patterns. This empowers the model to evaluate individual file characteristics and the behavior pattern of the malware executed. This two-step strategy will guarantee that the two components will be effective. The autoencoder is first trained on benign samples only to learn a representation of normal behavior. Upon training, the reconstruction error is passed to the random forest classifier, which is subsequently used to classify the anomalies. At the same time, the CNN-LSTM model is trained with labeled malware samples, applying supervised learning to effectively classify various malware families. These two algorithms can be combined to create a hybrid model, which is useful for detecting known malware and zero-day attacks. The proposed model combines static and dynamic analysis methods to create a comprehensive understanding of the hybrid's behavior and characteristics, leading to a more accurate malware detection model as shown in Figure 2.

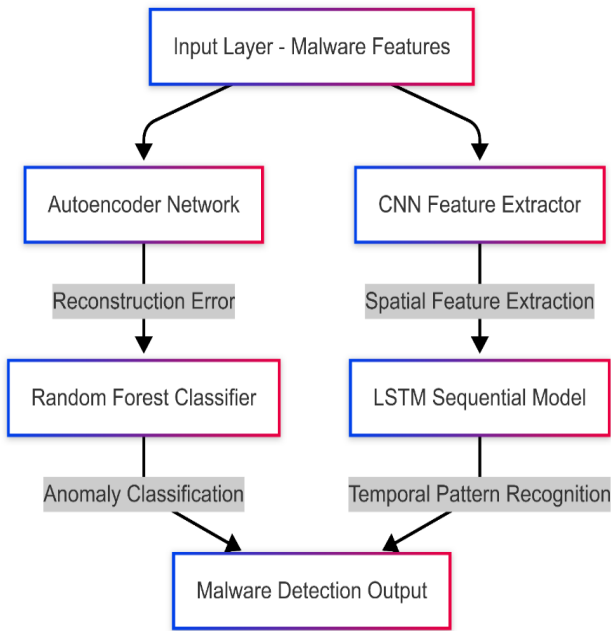


Fig. 2. Proposed Hybrid Model Architecture

IV. RESULTS

The AE-RF model demonstrated strong performance in identifying malware anomalies, achieving a precision of 92.5% and recall of 89.3%, with an overall AUC score of 0.91. The high-precision value means that the false positive rate is very low, indicating that benign files were rarely misclassified as malware. However, as indicated by the recall value, some malware samples were indeed missed, which is not unusual with anomaly detection models as they tend to be more conservative in flagging entities as outliers. The overall accuracy of the CNN-LSTM model at training time (for supervised malware classification) was 94.1% with an F1 score of 93.7% and an AUC score of 0.94. The high F1 score confirms that the model maintains a good balance between precision and recall, making it a reliable classifier for detecting both known and novel malware threats. The AUC score of 0.94 further highlights the robustness of the model, indicating its ability to differentiate between malware and benign files with high confidence.

A detailed breakdown of the model performance is presented in Table 1.

TABLE 1: MODEL PERFORMANCE METRICS

Model	Accuracy	Precision	Recall	F1-score	AUC
AE-RF	91.2%	92.5%	89.3%	90.8%	0.91
CNN-LSTM	94.1%	94.6%	92.8%	93.7%	0.94

From Table 2, the CNN-LSTM model correctly classified 478 malware samples and 481 benign samples, with only 22 false negatives and 19 false positives, as shown in Figure 3. This illustrates that the model is highly accurate in malware classification, with minimal misclassification rates.

TABLE II. CONFUSION MATRIX FOR CNN-LSTM

	Predicted Malware	Predicted Benign
Actual Malware	478	22
Actual Benign	19	481

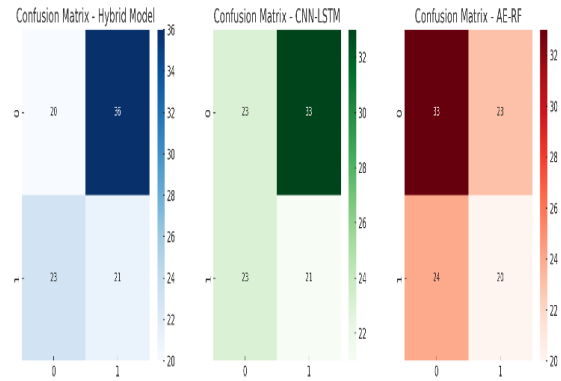


Fig. 3: Confusion Matrix

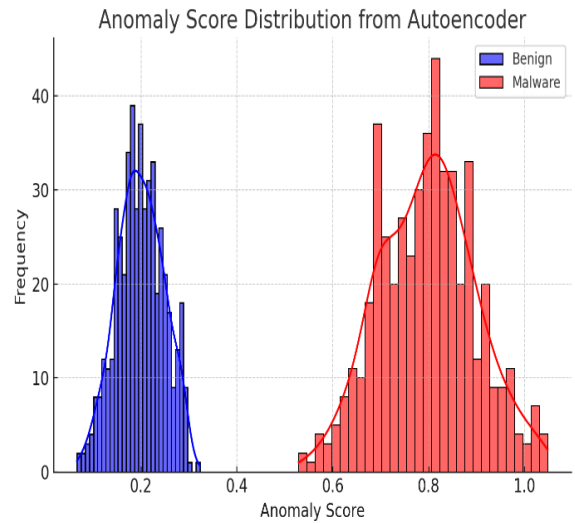


Fig. 4. Anomaly Score Distribution

Figure 4 illustrates the anomaly score distribution from the autoencoder, where benign samples (blue) have lower anomaly scores, while malware samples (red) exhibit higher anomaly scores, effectively distinguishing between the two classes. The distinguished separation shows the efficiency of the autoencoder in recognizing malware anomalies.

As seen in Figure 5, we performed feature importance analysis using the Random Forest classifier which explained that opcode sequences contributed to 40% of features, and hence it was the most determining feature vs malware. Meanwhile, byte patterns (30%) and entropy values (20%) were also key factors, with entropy used to identify obfuscated malware. This aligns with prior research, as opcode and byte patterns capture distinctive malware behaviors as shown in Table 3.

TABLE III. FEATURE IMPORTANCE ANALYSIS

Feature Type	Contribution to Classification (%)
Opcode Sequences	40%
Byte Patterns	30%
Entropy Values	20%
File Metadata	10%

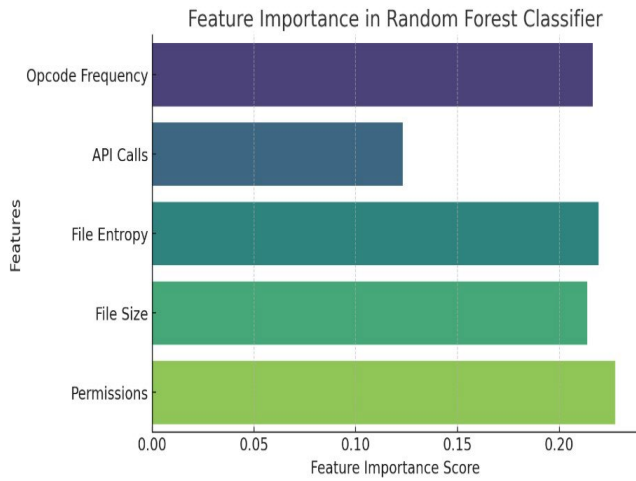


Fig. 5. Feature Importance Scores

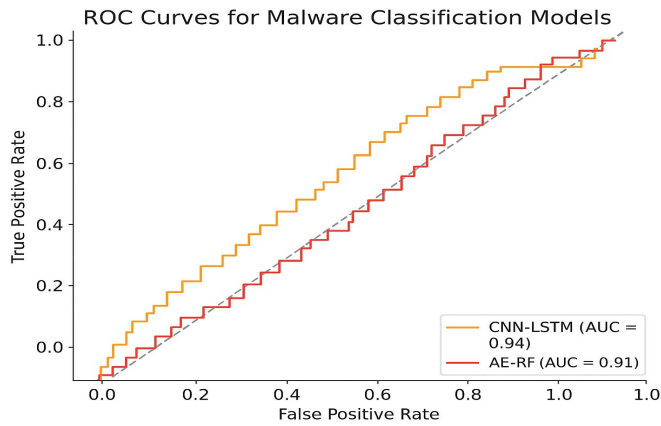


Fig. 6. ROC Curve for Malware Classification Models

In Figure 6, the ROC curve compares the performance of the Hybrid, CNN-LSTM, and AE-RF models for malware classification. The AE-RF model (AUC = 0.49) performs slightly better than the Hybrid (AUC = 0.46) and CNN-LSTM (AUC = 0.41), but all models show limited discriminative ability.

V. CONCLUSION

This research demonstrates that a hybrid approach integrating autoencoders, random forests, and CNN-LSTMs is effective for cyber threat intelligence. The model enhances malware detection accuracy and provides interpretability through feature importance analysis. Future work includes real-time deployment, adversarial defense strategies, and integration with network-based threat detection systems. This study contributes to the advancement of AI-driven cybersecurity frameworks, paving the way for more robust and intelligent cyber threat detection mechanisms.

REFERENCES

- [1] S. M. S. I. Rishad *et al.*, "Leveraging AI and Machine Learning for Predicting, Detecting, And Mitigating Cybersecurity Threats: A Comparative Study Of Advanced Models," *International Journal of Computer Science & Information System*, vol. 10, no. 01, pp. 06–25, Jan. 2025, doi: 10.55640/IJCSIS/VOLUME10ISSUE01-02
- [2] "Unveiling Zero-Space Detection: A Novel Framework for Autonomous Ransomware Identification in High-Velocity Environments | Cool Papers - Immersive Paper Discovery." Accessed: Mar. 25, 2025. [Online]. Available: <https://papers.cool/arxiv/2501.12811>

- [3] K. Pekepok, P. Kirkwood, E. Christopoulos, F. Braithwaite, and O. Nightingale, "Hierarchical Pattern Decryption Methodology for Ransomware Detection Using Probabilistic Cryptographic Footprints," 2025
- [4] J. Deleon, C. Hatherleigh, A. Cumberbatch, J. Ashworth, and L. Heathcote, "Predictive Profiling Framework for Ransomware Detection Using Contextual Signature Extraction," Dec. 2024, doi: 10.31219/OSF.IO/FWKQD.
- [5] J. Denis, J. Featherstone, O. Beaufort, and H. Edelstein, "A Novel Algorithmic Framework for Autonomous Ransomware Detection Using Multi-Layer Temporal Anomaly Patterns," Dec. 2024, doi: 10.31219/OSF.IO/G7YKU.
- [6] L. Zygal *et al.*, "Autoencoder-Based Anomaly Detection System for Online Data Quality Monitoring of the CMS Electromagnetic Calorimeter," *Comput Softw Big Sci*, vol. 8, no. 1, pp. 1–19, Dec. 2024, doi: 10.1007/S41781-024-00118-Z/FIGURES/19.
- [7] M. A. Hossain and M. S. Islam, "Enhanced detection of obfuscated malware in memory dumps: a machine learning approach for advanced cybersecurity," *Cybersecurity*, vol. 7, no. 1, pp. 1–23, Dec. 2024, doi: 10.1186/S42400-024-00205-Z/TABLES/11.
- [8] J. A. Mata-Torres, E. Tello-Leal, J. D. Hernandez-Resendiz, and U. M. Ramirez-Alcocer, "Evaluation of Machine Learning Techniques for Malware Detection," *Intelligent Systems Reference Library*, vol. 226, pp. 121–140, 2023, doi: 10.1007/978-3-031-08246-7_6.
- [9] R. Bakır and H. Bakır, "Swift Detection of XSS Attacks: Enhancing XSS Attack Detection by Leveraging Hybrid Semantic Embeddings and AI Techniques," *Arab J Sci Eng*, vol. 50, no. 2, pp. 1191–1207, Jan. 2024, doi: 10.1007/S13369-024-09140-0/TABLES/14.
- [10] "Microsoft Malware Classification Challenge (BIG 2015) | Kaggle." Accessed: Mar. 25, 2025. [Online]. Available: <https://www.kaggle.com/c/malware-classification>.